

File Permissions in Linux

Project Description

In this project, I reviewed and modified file and directory permissions within a Linux file system to ensure users had the correct level of authorisation. Using Linux commands, I looked at existing permissions, identified unauthorised access, and updated permissions to improve overall system security. This activity demonstrates foundational Linux administration and access control skills commonly used in cybersecurity environments.

Check File and Directory Details

To review the permissions of files and directories, including hidden files, I used the following command:

```
researcher2@2c8657b90227:~/projects$ ls -la
```

This command displays:

- File and directory permissions
- Hidden files
- File ownership
- Group ownership
- File sizes
- Modification dates

Example output:

```
drwxr-xr-x 3 researcher2 research_team 4096 May 21 19:20 .
drwxr-xr-x 3 researcher2 research_team 4096 May 21 19:46 ..
-rw--w---- 1 researcher2 research_team  46 May 21 19:20 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 May 21 19:20 drafts
-rw-rw-rw- 1 researcher2 research_team  46 May 21 19:20 project_k.txt
-rw-r----- 1 researcher2 research_team  46 May 21 19:20 project_m.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 21 19:20 project_r.txt
-rw-rw-r-- 1 researcher2 research_team  46 May 21 19:20 project_t.txt
researcher2@2c8657b90227:~/projects$
```

The current permissions showed that some of the files allowed unauthorised write access, which violated the organisation's security policy.

Describe the Permissions String

One example of a permission string is:

```
-rw-rw-r--
```

This 10-character string shows the permissions assigned to a file.

Permission explained:

- The first character indicated the file type
 - '-' means regular file
 - 'd' means directory

The remaining characters are divided into three groups:

- User permissions
- Group permissions
- Other permissions

Example:

```
-rw-rw-r--
```

- 'rw-' = user has read and write permissions
- 'rw-' = group has read and write permissions

- 'r-' = others have read-only permissions

Permission characters:

- 'r' = read
- 'w' = write
- 'x' = execute
- '-' = permission not granted

Change File Permissions

The organisation does not allow others to have write access to files. The file '*project_k.txt*' allowed write access for others and needed to be modified.

I used the following command:

```
researcher2@2c8657b90227:~/projects$ chmod o-w project_k.txt
```

Explanation:

- 'chmod' changes file permissions
- 'o' refers to others
- '-w' removes write permissions

After running the command, the permissions changed from:

```
-rw-rw-rw- 1 researcher2 research_team 46 May 21 19:20 project_k.txt
```

to:

```
-rw-rw-r-- 1 researcher2 research team 46 May 21 19:20 project k.txt
```

This removed unauthorised write access for other users.

Change File Permissions on a Hidden File

The hidden file `‘.project_x.txt’` should allow:

- user = read and write
- group = read only
- others = no access

I used the following command:

```
researcher2@2c8657b90227:~/projects$ chmod u=rw,g=r,o= .project_x.txt
```

Explanation:

- `‘u=rw’` gives the user read and write permissions
- `‘g=r’` gives the group read-only permissions
- `‘o=’` removes all permissions for others

This ensured the hidden file followed the organisation’s authorisation policy.

Change Directory Permissions

The `‘drafts’` directory should only be accessible by the `‘researcher2’` user.

I used the following command:

```
researcher2@2c8657b90227:~/projects$ chmod 700 drafts
```

Explanation:

- `‘7’` gives the user read, write, and execute permissions
- `‘0’` removes all permission from the group
- `‘0’` removes all permissions from others

This restricted access to the directory and its contents to only the authorised owner.

Summary

In this project, I used Linux commands to review and modify file and directory permissions within a file system. I used '`ls -la`' to inspect permissions and '`chmod`' to remove unauthorised access and apply secure authorisation settings. This activity strengthened my understanding of Linux file permissions, hidden files, access control, and least privilege principles used in cybersecurity.